Lehrstuhl Softwaretechnik & Programmiersprachen
Fakultät Wirtschaftsinformatik & Angewandte Informatik

SWT-FSE-B: Foundations of Software Engineering

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Exercise Sheet 7: PetriNets

7.1 Elementary Nets

Exercise 7.1.1 Model the control flow of a prototypical Login mechanism of the *SWT Library* System as an elementary net. The Login mechanism works as follows:

"A user opens the webpage of the SWT Library *System in a browser and is sent to the login page. When the user performs a login, they are sent to the overview page. It is assumed that every user opening the webpage will always login, and that each login attempt is successful. Afterwards, the user can logout and be sent to the login page. Finally, the user closes the browser. Due to hardware constraints, the number of logged-in users shall never exceed two. However, the number of users on the login page is unlimited."*

Exercise 7.1.2 Provide a run of the elementary net of Exercise 7.1.1 that correctly represents the following situation:

"A user opens the login page and logs in. Another user opens the login page and logs into the system. Both users logout. Finally, both users close their browser."

Exercise 7.1.3 Due to further hardware constraints, only a maximum of two users can now visit the webpage (login page and overview page) simultaneously. Provide a counterexample run showing that this new constraint is not fulfilled by the elementary net of Exercise 7.1.1. Adapt your previously proposed solution with respect to the new constraint.

Exercise 7.1.4 Prove that your proposed solution to Exercise 7.1.3 never reaches a marking that violates the following property:

"A maximum of two users can visit the webpage simultaneously."

Exercise 7.1.5 Briefly justify whether the Login mechanism can still be modelled with an elementary net if it were to be extended by the following functionality:

"When a user attempts to login with incorrect login credentials, they remain on the login page but are able to try and login again."

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7.2 UML Diagram Transformation

Exercise 7.2.1 Apply the normalization and translation rules presented in the lectures to translate the activity diagram of Figure 1 to an elementary net.

Exercise 7.2.2 Apply the translation rules presented in the lectures to translate the sequence diagram of Figure 2 to an elementary net. Additionally, propose a new translation rule for a self call (cf. *storeRequest* and *rollback*).

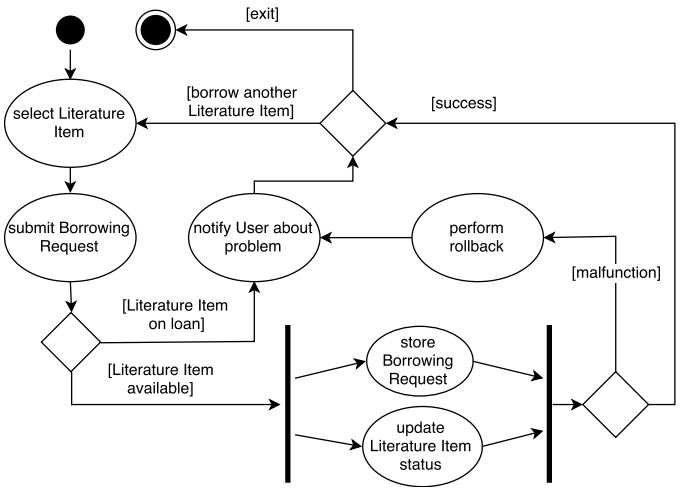


Figure 1: Adapted activity diagram of the *Borrowing a Literature Item* use case.

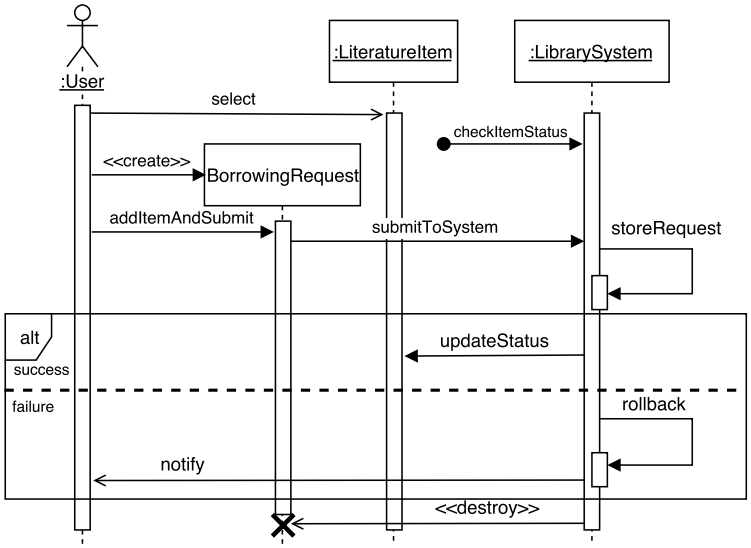


Figure 2: Adapted sequence diagram of the *Borrowing a Literature Item* use case.

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